Jiaqing Xie

Research Goals

I develop machine learning methods from theoretical view, with current interests in flow matching and diffusion models, graph neural networks and graph transformers, and applications in physical science. My work spans molecular and materials modeling, spectroscopy, protein learning, and scientific benchmarks.

Education

Fudan University 2026 – 2030
Ph.D. in Computer Science.

ETH Zurich 2022 – 2025
M.Sc. in Computer Science.

University of Edinburgh 2019 – 2022
B.Eng. in Electronics and Computer Science.

Research Interests

Flow matching and diffusion models.

Graph neural networks, graph transformers, positional encodings, and graph representation learning.

AI for physical science, including molecular modeling, spectroscopy, protein learning, chemistry agents, and materials benchmarks.

Honors & Awards

CCHO Regional Second Price Medal, 2016.

Publications

‡ indicates equal contribution.

Graph Neural Networks

Gauge-Invariant Learnable Spectral Positional Encodings for Directed Graphs via Hermitian Block Krylov Subspaces.
Jiaqing Xie, and Yuxin Wang. *To be released*.

Benchmarking Positional Encodings for GNNs and Graph Transformers.
Florian Grotschla‡, **Jiaqing Xie**‡, and Roger Wattenhofer. *SIGKDD 2026*.

AI for Physical Science

Antibody Design via On-Policy Self-Distillation.
Zhuo Yang, Jiaying He, **Jiaqing Xie**, et al.. *To be released*.

SpectraFlow: Physics-guided Flow Matching for IR-Raman Spectral Interconversion.
Jiaqing Xie, Zhuo Yang, Ben Gao, Shuaike Shen, Tianfan Fu, and Yuqiang Li. *To be released*.

NMRGym: A Comprehensive Benchmark for Nuclear Magnetic Resonance Based Molecular Structure Elucidation.
Zheng Fang, Chen Yang, Hai-tao Yu, Haoming Luo, Haitao He, **Jiaqing Xie**, Zhuo Yang, Yuqiang Li, and Jun Xia. *SIGKDD 2026*.

NMRTrans: Structure Elucidation from Experimental NMR Spectra via Set Transformers.
Liujia Yang‡, Zhuo Yang‡, **Jiaqing Xie**‡, Yubin Wang‡, Ben Gao, Tianfan Fu, Xingjian Wei, Jiaxing Sun, Jiang Wu, Conghui He, Yuqiang Li, and Qinying Gu. *SIGKDD 2026*.

SpectrumWorld: Artificial Intelligence Foundation for Spectroscopy.
Zhuo Yang‡, **Jiaqing Xie**‡, Shuaike Shen, Daolang Wang, Yeyun Chen, Ben Gao, Shuzhou Sun, Biqing Qi, Dongzhan Zhou, Lei Bai, Linjiang Chen, Shufei Zhang, Jun Jiang, Tianfan Fu, and Yuqiang Li. *SIGKDD 2026*.

PolyReal: A Benchmark for Real-World Polymer Science Workflows.
Wanhao Liu‡, Weida Wang‡, **Jiaqing Xie**, Suorong Yang, Jue Wang, Benteng Chen, Guangtao Mei, Zonglin Yang, Shufei Zhang, Yuchun Mo, Lang Cheng, Jin Zeng, Houqiang Li, Wanli Ouyang, and Yuqiang Li. *CVPR 2026 Findings*.

DeepProtein: Deep Learning Library and Benchmark for Protein Sequence Learning.
Jiaqing Xie, Yuqiang Li, and Tianfan Fu. *Bioinformatics 2025*.

Large Language Models

R2-Seg: Training-Free OOD Medical Tumor Segmentation via Anatomical Reasoning and Statistical Rejection.
Shuaike Shen‡, Ke Liu‡, **Jiaqing Xie**, Shangde Gao, Chunhua Shen, Ge Liu, Mireia Crispin-Ortuzar, and Shangqi Gao. *CVPR 2026, Best Paper Candidate*.

QCBench: Evaluating Large Language Models on Domain-Specific Quantitative Chemistry.
Jiaqing Xie‡, Weida Wang‡, Ben Gao‡, Zhuo Yang, Haiyuan Wan, Shufei Zhang, Tianfan Fu, and Yuqiang Li. *JCIM 2025*.

Deep Research Arena: The First Exam of LLMs' Research Abilities via Seminar-Grounded Tasks.
Haiyuan Wan, Chen Yang, Junchi Yu, Meiqi Tu, Jiaxuan Lu, Di Yu, Jianbao Cao, Ben Gao, **Jiaqing Xie**, Aoran Wang, Wenlong Zhang, Philip Torr, and Dongzhan Zhou. *AAAI 2026*.

Technical Reports and Under Review

MolAct: An Agentic RL Framework for Molecular Editing and Property Optimization.
Zhuo Yang‡, Yeyun Chen‡, **Jiaqing Xie**, Ben Gao, Shuaike Shen, Wanhao Liu, Liujia Yang, Beilun Wang, Tianfan Fu, and Yuqiang Li. *Under Review, arXiv*.

A Comparative Analysis of Sparse Autoencoder and Activation Difference in Language Model Steering.
Jiaqing Xie. *Under Review, arXiv*.

GeoBA: Geometry-Biased Attention for Molecular Property Prediction.
Jiaqing Xie, Ziheng Chi, and Zhuo Yang. *Under Review, arXiv*.

SkillsInjector: Dynamic Skill Context Construction for LLM Agents.
Yanchao Li, Wanhao Liu, Ben Gao, **Jiaqing Xie**, Zhehong Ai, Na Zou, Yuqiang Li, and Tianfan Fu. *Under Review, arXiv*.

OmniMatBench: A Human-Calibrated Multimodal Reasoning Benchmark Across 19 Materials Science Subfields.
Wanhao Liu‡, **Jiaqing Xie**‡, Qian Tan, Weida Wang, Jue Wang, Ran Sun, Zhuo Yang, Lei Bai, Tianfan Fu, Lu Chen, Xin Chen, and Yuqiang Li. *Under Review, arXiv*.

SpecMol: A Unified Framework for Spectroscopy-Grounded Molecular Modeling and Evaluation.
Shuaike Shen‡, **Jiaqing Xie**‡, Zhuo Yang, Antong Zhang, Shuzhou Sun, Ben Gao, Tianfan Fu, Biqing Qi, and Yuqiang Li. *Under Review, arXiv*.

Selected Research Projects

DeepProtein – Deep learning library and benchmark for protein sequence learning. Developed a benchmark-oriented software and research project for protein learning; published in *Bioinformatics* 2025.

QCBench – LLM evaluation for quantitative chemistry. Built domain-specific evaluation tasks for quantitative chemistry and scientific reasoning; published in *Journal of Chemical Information and Modeling* 2025.

GraphPEBench – Graph positional encodings. Co-developed benchmarking work on positional encodings for GNNs and graph transformers; accepted to SIGKDD 2026.

NMRGym and NMRTrans – NMR-based molecular structure elucidation. Worked on benchmark construction and set-transformer modeling for NMR spectroscopy; accepted to SIGKDD 2026.

SpectrumWorld and SpectraFlow – AI foundation and flow matching for spectroscopy. Developed spectroscopy-centered models and benchmarks for IR-Raman conversion and broader molecular/spectral reasoning.

ChemClaw and GeoBA – Chemistry agents and molecular property prediction. Current projects on agentic chemistry workflows and geometry-biased attention for molecular representation learning.

Academic Service

Reviewer, 2026: NeurIPS

Reviewer, 2025: ACL, CVPR, EMNLP, ICONIP, NeurIPS.

Reviewer, 2024: ACML, NeurIPS.

Technical Skills

Programming and tools: Python, PyTorch, PyTorch Geometric, e3nn, RDKit, Linux, Git, and reproducible ML workflows.

Additional Background

Weiqi / Go: Chinese Weiqi Association amateur 1 dan 2009.

Programming contests: Codeforces 1200.

Activities: running and singing.

References

Available upon request.